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ADDIS ABABA UNIVERSITY

ADDIS ABABA INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING

Department of [Name]

Title of the Thesis: Concise, Clear and Descriptive

A thesis submitted to the School of Information Technology and Engineering, Addis Ababa Institute of Technology, Addis Ababa University, in partial fulfillment of the requirements for the degree of Master of Science in [Program Name]

By

Full Name of the Student

Advisor: Dr. Full Name of the Advisor

Co-advisor: Dr. Full Name of the Co-advisor

Addis Ababa, Ethiopia

November 2026

ADDIS ABABA UNIVERSITY

ADDIS ABABA INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING

Title of the Thesis: Concise, Clear and Descriptive

By

Full Name of the Student

Approval by Board of Examiners

This is to certify that the thesis prepared by Full Name of the Student, entitled "Title of the Thesis: Concise, Clear and Descriptive", and submitted in partial fulfillment of the requirements for the degree of Master of Science in [Program Name], complies with the regulations of the University and meets the accepted standards with respect to originality and quality.

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Advisor	Dr. Full Name of the Advisor	_____	_____
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External Examiner	_____	_____	_____
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DECLARATION

I, the undersigned, declare that this thesis is my original work, that it has not been presented for a degree in this or any other university, and that all sources of material used for the thesis have been duly acknowledged.

	Name	**Signature**	**Date**
Student	Full Name of the Student	_____	_____

This thesis has been submitted for examination with my approval as university advisor.

	Name	**Signature**	**Date**
Advisor	Dr. Full Name of the Advisor	_____	_____
Co-advisor	Dr. Full Name of the Co-advisor	_____	_____

Place: Addis Ababa, Ethiopia
Date of submission: November 2026

To . . .

ABSTRACT

Guide: One page that tells the reader why the problem matters, what you did and how, what the main results were (with numbers), and what the work contributes. Write it last, and do not cite references in it.

Write the abstract here. Start with the problem and why it matters. Then describe the approach and datasets, give the main quantitative results, and end with the contribution of the work.

Keywords: keyword one, keyword two, keyword three, keyword four

ACKNOWLEDGMENT

Write your acknowledgment here.

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LIST OF ACRONYMS

AAiT	Addis Ababa Institute of Technology
AAU	Addis Ababa University
CNN	Convolutional Neural Network
ConvLSTM	Convolutional Long Short-Term Memory
GPU	Graphics Processing Unit
SiTE	School of Information Technology and Engineering

CHAPTER ONE

INTRODUCTION

1.1. Background

> **Guide:** Give the reader the facts they need to understand the topic and how it links to your research question. Start from the broad theme, then narrow down to the specific details that make your problem relevant. By the end, the need for the research should already be clear.

Write the background here. Cite sources as you go [1], and introduce acronyms with `\ac{...}`. For example, Convolutional Neural Network (CNN) is spelled out on first use and abbreviated afterwards: CNN.

1.2. Motivation

> **Guide:** Explain what motivated you to carry out this work.

Write the motivation here.

1.3. Statement of the Problem

> **Guide:** Define the problem in the literature, theory or practice that this thesis solves, clearly enough that the reader knows exactly what is being addressed. A good problem statement answers "Why does this research need to be conducted?"

Write the problem statement here.

1.4. Research Questions

> **Guide:** List the questions the thesis answers. Chapter 6 should answer each one explicitly.

This study answers the following research questions:

RQ1. First research question?

RQ2. Second research question?

RQ3. Third research question?

1.5. Objectives

Guide: State a general objective, and specific objectives that are simple, specific and follow from the problem statement.

1.5.1. General Objective

The general objective of this study is to . . .

1.5.2. Specific Objectives

To achieve the general objective, the study will:

- review . . . ;
- prepare . . . ;
- design and implement . . . ;
- evaluate . . . .

1.6. Research Methodology

Guide: A short overview of how the objectives were achieved: literature review, data, tools and evaluation. Keep it brief and point to Chapter 4 for the details.

1.6.1. Literature Review

Describe how the literature was reviewed.

1.6.2. Data Collection

Describe the dataset(s) used.

1.6.3. Tools and Techniques

List the main tools, frameworks and hardware.

1.6.4. Evaluation

Describe how the results were evaluated.

1.7. Scope and Limitations

Guide: State what the study covers, and what it does not cover and why.

Scope. Describe what the study covers.

Limitations. Describe what the study does not cover.

1.8. Contributions

Guide: List the concrete contributions of the thesis (new method, dataset, results, analysis) and how they extend existing knowledge.

The main contributions of this thesis are:

- . . . ;
- . . . .

1.9. Organization of the Thesis

The rest of this thesis is organized as follows. Chapter 2 reviews the background concepts and techniques. Chapter 3 discusses related work and identifies the research gap. Chapter 4 presents the proposed approach. Chapter 5 describes the experiments and discusses the results. Chapter 6 concludes the thesis and outlines future work.

CHAPTER TWO

LITERATURE REVIEW

Guide: Cover the concepts, theories and techniques the reader needs to understand the rest of the thesis. Organise the chapter by topic, cite the original sources, and keep it focused on what your work builds on.

Give a short introduction to the chapter here.

2.1. Overview of [Topic]

Write about the topic here. A book is cited like this [2].

2.2. [Core Concept or Technique]

2.2.1. [Sub-topic]

Write about the sub-topic here.

2.2.2. [Sub-topic]

Write about the sub-topic here.

2.3. Datasets

Describe the publicly available datasets in the area.

2.4. Evaluation Metrics

Describe the metrics commonly used in the area [3].

2.5. Summary

Summarise the chapter in a short paragraph.

CHAPTER THREE

RELATED WORK

Guide: Discuss the studies closest to your work, grouped by approach. For each group, say what was done, how well it worked and what is still missing. End by stating the gap that this thesis fills, and show how your work is similar to, yet different from, previous work.

Give a short introduction to the chapter here.

3.1. [First Group of Approaches]

Discuss the first group of related works here [4].

3.2. [Second Group of Approaches]

Discuss the second group of related works here [5].

3.3. Summary and Research Gap

Every table must be referred to in the text. Table 3.1 summarises the related works discussed in this chapter.

Table 3.1: Summary of related works

Work	Approach	Dataset / Evaluation	Limitation / Gap
[1]	...	...	...
[4]	...	...	...
[5]	...	...	...

State the research gap that this thesis addresses.

CHAPTER FOUR

METHODOLOGY

> **Guide:** Describe what you built or did, in enough detail that someone else could reproduce it. Justify each design choice and relate it back to the research questions.

Give a short introduction to the chapter here.

4.1. Research Design

Describe the overall research approach and why it suits the research questions.

4.2. Dataset

4.2.1. Data Collection

Describe where the data comes from, its size and how it is split.

4.2.2. Preprocessing

Describe how the data is cleaned and prepared.

4.3. Proposed Architecture

Every figure must be referred to in the text. Figure 4.1 shows the overall architecture of the proposed approach.

4.3.1. [Component One]

Describe the first component.

4.3.2. [Component Two]

Describe the second component. Algorithm 4.1 shows how an algorithm is written.

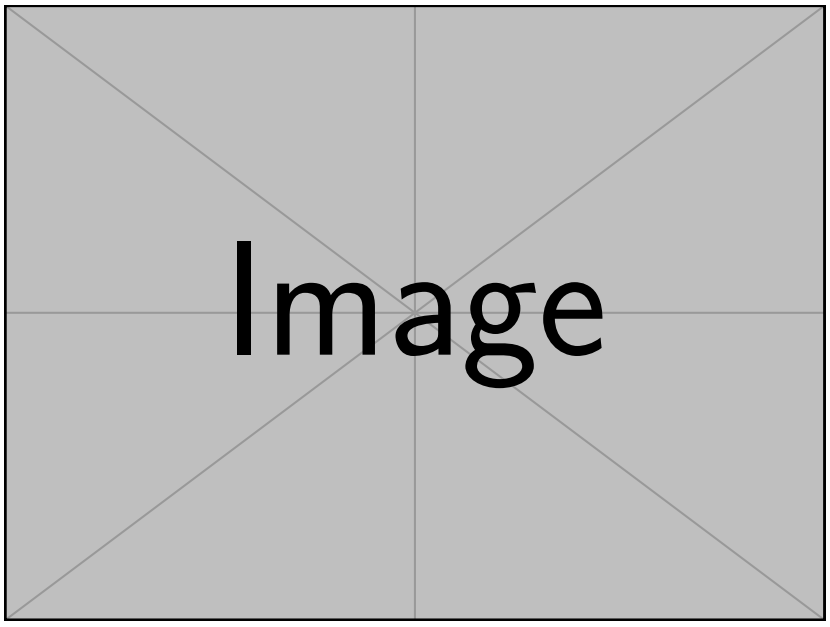


Figure 4.1: Overall architecture of the proposed approach

Algorithm 4.1 Example algorithm

Require: input x
Ensure: output y
1: $y \leftarrow 0$
2: **for** $i \leftarrow 1$ **to** n **do**
3: $\quad y \leftarrow y + f(x_i)$
4: **end for**
5: **return** y

4.4. Training Procedure

Describe the loss function, optimiser and training strategy. Number equations so they can be referred to, e.g. Equation (4.1):

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \ell \left(\hat{y}_i, y_i \right) . \tag{4.1}$$

4.5. Evaluation Metrics

Define the metrics used to evaluate the approach and explain why they were chosen.

4.6. Ethical Considerations

Discuss any ethical issues, for example the privacy of people in the data.

CHAPTER FIVE

EXPERIMENTS, RESULTS AND DISCUSSION

Guide: Describe the setup, present the results objectively, compare them with existing methods, and then interpret them. The discussion should say what the results mean for each research question.

Give a short introduction to the chapter here.

5.1. Experimental Setup

Table 5.1 lists the hardware and software used in the experiments.

Table 5.1: Experimental environment

Item	Specification
Graphics Processing Unit (GPU)	...
CPU / RAM	...
Operating system	...
Framework	...

5.1.1. Implementation Details

Give the hyperparameters and other details needed to reproduce the results.

5.2. Results

5.2.1. Quantitative Results

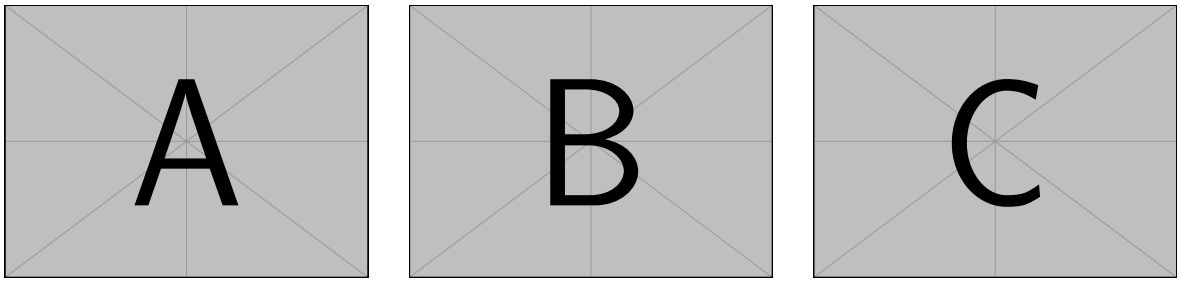
Table 5.2 compares the proposed approach with existing methods. Highlight the best result in each column in bold.

Table 5.2: Comparison with existing methods

Method	Metric 1 ↑	Metric 2 ↑	Metric 3 ↓
Method A [4]	0.000	0.000	0.000
Method B [5]	0.000	0.000	0.000
Proposed	**0.000**	**0.000**	**0.000**

5.2.2. Qualitative Results

Figure 5.1 shows example outputs.



(a) Input **(b)** Ground truth **(c)** Proposed

Figure 5.1: Qualitative results

5.3. Ablation Study

Show the effect of each component of the proposed approach.

5.4. Discussion

Interpret the results and answer each research question from Section 1.4.

CHAPTER SIX

CONCLUSION AND FUTURE WORK

6.1. Conclusion

> **Guide:** Summarise the problem, what was done and the main findings, and answer each research question briefly. Do not introduce new results here.

Write the conclusion here.

6.2. Contributions

Restate the main contributions of the thesis.

6.3. Future Work

> **Guide:** Describe the limitations of the work and the directions that follow from them.

Describe future work here.

REFERENCES

[1] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 20, no. 11, pp. 1254–1259, 1998. DOI: 10.1109/34.730558.

[2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.

[3] Z. Bylinskii, T. Judd, A. Oliva, A. Torralba, and F. Durand, "What do different evaluation metrics tell us about saliency models?" *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 3, pp. 740–757, 2019. DOI: 10.1109/TPAMI.2018.2815601.

[4] W. Wang, J. Shen, F. Guo, M.-M. Cheng, and A. Borji, "Revisiting video saliency: A large-scale benchmark and a new model," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4894–4903.

[5] K. Min and J. J. Corso, "TASED-Net: Temporally-aggregating spatial encoder-decoder network for video saliency detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

APPENDIX A

ADDITIONAL RESULTS

Put additional tables and figures here.

APPENDIX B

SAMPLE CODE

Put important code listings or sample outputs here.